

1. Create one VPC in N. Virginia region.

The screenshot shows the AWS VPC console for a VPC named 'vpc-02bf85d4183f5fd76 / Autoscaling-VPC'. A green banner at the top states 'You successfully created vpc-02bf85d4183f5fd76 / Autoscaling-VPC'. The left sidebar shows the 'VPC dashboard' with a filter set to 'Your VPCs'. The main content area displays the VPC details in a grid:

- VPC ID:** vpc-02bf85d4183f5fd76
- State:** Available
- Block Public Access:** Off
- DNS hostnames:** Disabled
- DNS resolution:** Enabled
- Tenancy:** default
- DHCP option set:** dopt-0fcb7e541a882ba45
- Main network ACL:** acl-056be63a74303710c
- Default VPC:** No
- IPv4 CIDR:** 10.0.0.0/16
- Main route table:** rtb-08d0092b5dbcfded4
- IPv6 CIDR (Network border group):** -
- Network Address Usage metrics:** Disabled
- Route 53 Resolver DNS Firewall rule groups:** -
- Encryption control ID:** -
- Encryption control mode:** -
- Owner ID:** 956035085469

Below the details is a 'Resource map' section showing the VPC, Subnets (0), and Route tables (1).

2. Create two subnets: one public subnet and one private subnet.

The screenshot shows the AWS VPC console for the 'Subnets' page. A green banner at the top states 'You successfully created subnet-09f272cca8d5b6feda / Autoscaling-Subnet'. The left sidebar shows the 'VPC dashboard' with a filter set to 'Your VPCs'. The main content area displays a table of subnets:

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
Public-Subnet	subnet-09f272cca8d5b6feda	Available	vpc-02bf85d4183f5fd76 Autoscaling-VPC	Off	10.0.1.0/24
Private-Subnet	subnet-0c5b3434319b8b03d	Available	vpc-02bf85d4183f5fd76 Autoscaling-VPC	Off	10.0.2.0/24

Below the table is a 'Select a subnet' section.

3. Attach an IGW to the VPC.

The screenshot shows the AWS VPC console for the 'Internet gateways' page. A green banner at the top states 'You successfully created igw-00954b6f6207594fe / Autoscaling-IGW'. The left sidebar shows the 'VPC dashboard' with a filter set to 'Your VPCs'. The main content area displays a table of internet gateways:

Name	Internet gateway ID	State	VPC ID	Owner
Autoscaling-IGW	igw-00954b6f6207594fe	Attached	vpc-02bf85d4183f5fd76 Autoscaling-VPC	956035085469

Below the table is a 'Select an internet gateway above' section.

4. Create one public route table (RT) and one private route table.

aws

Search

[Alt+S]

United States (N. Virginia)

Poojith R (956035085469)

VPC > Route tables > rtb-084fbef6d03449012

Updated routes for rtb-084fbef6d03449012 / Public-RT successfully
Details

rtb-084fbef6d03449012 / Public-RT

Actions

VPC dashboard

Filter by VPC:

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Details

Info

Route table ID

rtb-084fbef6d03449012

VPC

vpc-02bf85d4183f5fd76 | Autoscaling-VPC

Main

No

Owner ID

956035085469

Explicit subnet associations

-

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (2)

Filter routes

Both

Edit routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-00954b6f6207594fe	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

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United States (N. Virginia)

Poojith R (956035085469)

VPC > Route tables > rtb-08d0092b5dbcfdded4

rtb-08d0092b5dbcfdded4 / Private-RT

Actions

VPC dashboard

Filter by VPC:

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Details

Info

Route table ID

rtb-08d0092b5dbcfdded4

VPC

vpc-02bf85d4183f5fd76 | Autoscaling-VPC

Main

Yes

Owner ID

956035085469

Explicit subnet associations

subnet-0c5b3434319b8b03d / Private-Subnet

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (1)

Filter routes

Both

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table

aws

Search

[Alt+S]

United States (N. Virginia)

Poojith R (956035085469)

VPC > Route tables

Route tables (3)

Last updated less than a minute ago

Actions

Create route table

VPC dashboard

Filter by VPC:

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	-	rtb-0d668f9c7b98dcb2	-	-	Yes	vpc-0f3216a7937311169
☐	Private-RT	rtb-08d0092b5dbcfdded4	subnet-0c5b3434319b8b...	-	Yes	vpc-02bf85d4183f5fd76 Autos
☐	Public-RT	rtb-084fbef6d03449012	subnet-09f27cca8d5b6fe...	-	No	vpc-02bf85d4183f5fd76 Autos

5. Deploy a NAT gateway in the public subnet and attach the NAT gateway to the private subnet.

The screenshot shows the AWS Management Console for a NAT gateway. The left sidebar contains the VPC dashboard menu. The main content area displays the details for the NAT gateway **nat-039d4f278812a1008 / Autoscaling-NAT**. The details include:

- NAT gateway ID:** nat-039d4f278812a1008
- NAT gateway ARN:** arn:aws:ec2:us-east-1:956035085469:natgateway/nat-039d4f278812a1008
- VPC:** vpc-02bf85d4183f5fd76 / Autoscaling-VPC
- Connectivity type:** Public
- Primary public IPv4 address:** 100.51.70.122
- Subnet:** subnet-09f27cca8d5b6feda / Public-Subnet
- State:** Available
- Primary private IPv4 address:** 10.0.1.118
- Created:** Wednesday, 16 September 2026 at 15:57:29 GMT+5:30
- State message:** Info
- Primary network interface ID:** eni-0515dc88eae835283
- Deleted:** -

Below the details, there are tabs for **Secondary IPv4 addresses**, **Monitoring**, and **Tags**. The **Secondary IPv4 addresses** tab is active, showing a table with columns: Private IPv4 address, Allocation ID, Association ID, and Public IPv4 address. A message states: "Secondary IPv4 addresses are not available for this nat gateway."

The screenshot shows the AWS Management Console for a route table. The left sidebar contains the VPC dashboard menu. The main content area displays the details for the route table **rtb-08d0092b5dbcfed4 / Private-RT**. The details include:

- Route table ID:** rtb-08d0092b5dbcfed4
- VPC:** vpc-02bf85d4183f5fd76 / Autoscaling-VPC
- Main:** Yes
- Owner ID:** 956035085469
- Explicit subnet associations:** subnet-0c5b3434319b8b03d / Private-Subnet
- Edge associations:** -

Below the details, there are tabs for **Routes**, **Subnet associations**, **Edge associations**, **Route propagation**, **Tags**, and **Related resources**. The **Routes** tab is active, showing a table with columns: Destination, Target, Status, Propagated, and Route Origin. The table contains two routes:

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-039d4f278812a1008	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

6. Create two instances, one in the public subnet and one in the private subnet.

The screenshot shows the AWS Management Console for EC2 instances. The left sidebar contains the EC2 dashboard menu. The main content area displays the details for two EC2 instances. The table shows the following information:

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Put
Public-Apache...	i-006a6a7d6d460316b	Running	t3.micro	Initializing	us-east-1a	-	44.
Private-Apach...	i-0370cef2dc3dae283	Running	t3.micro	Initializing	us-east-1a	-	-

7. Deploy Apache server on both EC2 instances with a sample index.html file.

```
ec2-user@ip-10-0-1-91 ~$ ssh -i Test1.pem ec2-user@44.204.198.24
Warning: Permanently added '44.204.198.24' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-0-1-91 ~]$ sudo dnf update -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-1-91 ~]$ sudo dnf install httpd -y
Last metadata expiration check: 0:00:14 ago on Wed Sep 16 10:42:01 2026.
Dependencies resolved.

Package                               Architecture      Version            Repository          Size
-----
Installing:
httpd                                x86_64            2.4.68-1.amzn2023.0.1  amazonlinux        46 k
Installing dependencies:
apr                                  x86_64            1.7.5-1.amzn2023.0.4  amazonlinux        129 k
apr-util                             x86_64            1.6.5-1.amzn2023.0.1  amazonlinux        100 k
apr-util-ldap                        x86_64            1.6.5-1.amzn2023.0.1  amazonlinux        15 k
generic-logos-httpd                 noarch            18.0.0-12.amzn2023.0.3  amazonlinux        19 k
httpd-core-2.4.68-1.amzn2023.0.1    x86_64            2.4.68-1.amzn2023.0.1  amazonlinux        1.4 M
httpd-filesystem                    noarch            2.4.68-1.amzn2023.0.1  amazonlinux        12 k
httpd-tools                          x86_64            2.4.68-1.amzn2023.0.1  amazonlinux        80 k
libbrotli                           x86_64            1.0.9-4.amzn2023.0.2  amazonlinux        315 k
mailcap                              noarch            2.1.49-3.amzn2023.0.3  amazonlinux        33 k
Installing weak dependencies:
apr-util-openssl                    x86_64            1.6.5-1.amzn2023.0.1  amazonlinux        17 k
mod_http2                           x86_64            2.0.42-1.amzn2023.0.1  amazonlinux        167 k
mod_lua                              x86_64            2.4.68-1.amzn2023.0.1  amazonlinux        59 k

Transaction Summary
Install 13 Packages

Total download size: 2.4 M
Installed size: 7.0 M
Downloading Packages:
(1/13): apr-util-1.6.5-1.amzn2023.0.1.x86_64.rpm                4.2 MB/s | 100 kB  00:00
(2/13): apr-util-ldap-1.6.5-1.amzn2023.0.1.x86_64.rpm           600 kB/s | 15 kB  00:00
(3/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm                     4.2 MB/s | 129 kB  00:00
(4/13): apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64.rpm        577 kB/s | 17 kB  00:00
(5/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm   643 kB/s | 19 kB  00:00
```

```
Installing      : libbrotli-1.0.9-4.amzn2023.0.2.x86_64
Running scriptlet: httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch
Installing      : httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch
Installing      : httpd-core-2.4.68-1.amzn2023.0.1.x86_64
Installing      : mod_http2-2.0.42-1.amzn2023.0.1.x86_64
Installing      : mod_lua-2.4.68-1.amzn2023.0.1.x86_64
Installing      : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
Installing      : httpd-2.4.68-1.amzn2023.0.1.x86_64
Running scriptlet: httpd-2.4.68-1.amzn2023.0.1.x86_64
Verifying       : apr-1.7.5-1.amzn2023.0.4.x86_64
Verifying       : apr-util-1.6.5-1.amzn2023.0.1.x86_64
Verifying       : apr-util-ldap-1.6.5-1.amzn2023.0.1.x86_64
Verifying       : apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64
Verifying       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
Verifying       : httpd-core-2.4.68-1.amzn2023.0.1.x86_64
Verifying       : httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch
Verifying       : httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch
Verifying       : httpd-tools-2.4.68-1.amzn2023.0.1.x86_64
Verifying       : libbrotli-1.0.9-4.amzn2023.0.2.x86_64
Verifying       : mailcap-2.1.49-3.amzn2023.0.3.noarch
Verifying       : mod_http2-2.0.42-1.amzn2023.0.1.x86_64
Verifying       : mod_lua-2.4.68-1.amzn2023.0.1.x86_64

apr-1.7.5-1.amzn2023.0.4.x86_64      apr-util-1.6.5-1.amzn2023.0.1.x86_64      apr-util-ldap-1.6.5-1.amzn2023.0.1.x86_64      apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      httpd-2.4.68-1.amzn2023.0.1.x86_64      httpd-core-2.4.68-1.amzn2023.0.1.x86_64      httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch
httpd-tools-2.4.68-1.amzn2023.0.1.x86_64      libbrotli-1.0.9-4.amzn2023.0.2.x86_64      mailcap-2.1.49-3.amzn2023.0.3.noarch      mod_http2-2.0.42-1.amzn2023.0.1.x86_64
mod_lua-2.4.68-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-10-0-1-91 ~]$ sudo systemctl start httpd
[ec2-user@ip-10-0-1-91 ~]$ sudo systemctl enable httpd
created symlink /etc/systemd/system/multi-user.target.wants/httpd.service - /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-10-0-1-91 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-09-16 10:42:28 UTC; 17s ago
     Docs: man:httpd.service(8)
  Main PID: 19265 (httpd)
    Status: 'Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:  0 B/sec'
    Tasks: 177 (limit: 1059)
   Memory: 13.7M
      CPU: 0m
   CGroup: /system.slice/httpd.service
           └─19265 /usr/sbin/httpd -DFOREGROUND
           └─19351 /usr/sbin/httpd -DFOREGROUND
           └─19353 /usr/sbin/httpd -DFOREGROUND
           └─19355 /usr/sbin/httpd -DFOREGROUND
           └─19357 /usr/sbin/httpd -DFOREGROUND
           └─19363 /usr/sbin/httpd -DFOREGROUND

Sep 16 10:42:28 ip-10-0-1-91.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Sep 16 10:42:28 ip-10-0-1-91.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Sep 16 10:42:28 ip-10-0-1-91.ec2.internal httpd[19265]: Server configured, listening on: port 80
[ec2-user@ip-10-0-1-91 ~]$ ssh -i /home/ec2-user/Test1.pem ec2-user@10.0.2.206
The authenticity of host '10.0.2.206 (10.0.2.206)' can't be established.
ED25519 key fingerprint is SHA256:kxvba8n3SgqCEQ98iVJ0N4eBRj9HxfDpRtXvF5a0s.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
warning: Permanently added '10.0.2.206' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-0-2-206 ~]$ hostname
ip-10-0-2-206.ec2.internal
```

Hello from Public Apache Server

```
[ec2-user@ip-10-0-1-91 ~]$ ls -l /home/ec2-user/Test1.pem
-rw-r--r-- 1 ec2-user ec2-user 1678 Sep 16 10:49 /home/ec2-user/Test1.pem
[ec2-user@ip-10-0-1-91 ~]$ chmod 400 /home/ec2-user/Test1.pem
[ec2-user@ip-10-0-1-91 ~]$ ssh -i /home/ec2-user/Test1.pem ec2-user@10.0.2.206
The authenticity of host '10.0.2.206 (10.0.2.206)' can't be established.
ED25519 key fingerprint is SHA256:kxvba8n3SgqCEQ98iVJ0N4eBRj9HxfDpRtXvF5a0s.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
warning: Permanently added '10.0.2.206' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-0-2-206 ~]$ hostname
ip-10-0-2-206.ec2.internal
```

```

[ec2-user@ip-10-0-2-206 ~]$ sudo dnf update -y
amazon linux 2023 kernel Livepatch repository                                     524 kB/s | 75 kB    00:00
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-10-0-2-206 ~]$ sudo dnf install httpd -y
Last metadata expiration check: 0:00:12 ago on Wed Sep 16 10:52:12 2026.
Dependencies resolved.

```

Package	Architecture	Version	Repository	Size
Installing:				
httpd	x86_64	2.4.68-1.amzn2023.0.1	amazonlinux	46 k
Installing dependencies:				
apr	x86_64	1.7.5-1.amzn2023.0.4	amazonlinux	129 k
apr-util	x86_64	1.6.5-1.amzn2023.0.1	amazonlinux	100 k
apr-util-ldb	x86_64	1.6.5-1.amzn2023.0.1	amazonlinux	15 k
generic-logos-httpd	noarch	18.0.0-12.amzn2023.0.3	amazonlinux	19 k
httpd-core	x86_64	2.4.68-1.amzn2023.0.1	amazonlinux	1.4 M
httpd-filesystem	noarch	2.4.68-1.amzn2023.0.1	amazonlinux	12 k
httpd-tools	x86_64	2.4.68-1.amzn2023.0.1	amazonlinux	80 k
libbrotli	x86_64	1.0.9-4.amzn2023.0.2	amazonlinux	315 k
mailcap	noarch	2.1.49-3.amzn2023.0.3	amazonlinux	33 k
Installing weak dependencies:				
apr-util-openssl	x86_64	1.6.5-1.amzn2023.0.1	amazonlinux	17 k
mod_http2	x86_64	2.0.42-1.amzn2023.0.1	amazonlinux	167 k
mod_lua	x86_64	2.4.68-1.amzn2023.0.1	amazonlinux	59 k

```

Transaction Summary
-----
Install 13 Packages

Total download size: 2.4 M
Installed size: 7.0 M
Downloading Packages:
(1/13): apr-util-ldb-1.6.5-1.amzn2023.0.1.x86_64.rpm                    501 kB/s | 15 kB    00:00
(2/13): apr-util-1.6.5-1.amzn2023.0.1.x86_64.rpm                     3.0 MB/s | 100 kB   00:00
(3/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm                          3.3 MB/s | 129 kB   00:00
(4/13): apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64.rpm             3.4 kB/s | 17 kB    00:00
(5/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm        620 kB/s | 19 kB    00:00
(6/13): httpd-2.4.68-1.amzn2023.0.1.x86_64.rpm                      1.5 MB/s | 46 kB    00:00
(7/13): httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch.rpm            440 kB/s | 12 kB    00:00
(8/13): httpd-core-2.4.68-1.amzn2023.0.1.x86_64.rpm                  32 MB/s | 1.4 MB   00:00
(9/13): httpd-tools-2.4.68-1.amzn2023.0.1.x86_64.rpm                 2.0 MB/s | 80 kB    00:00
(10/13): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm                    1.3 MB/s | 33 kB    00:00
(11/13): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm                   7.1 MB/s | 315 kB   00:00
(12/13): mod_http2-2.0.42-1.amzn2023.0.1.x86_64.rpm                  5.1 MB/s | 167 kB   00:00
(13/13): mod_lua-2.4.68-1.amzn2023.0.1.x86_64.rpm                    2.1 MB/s | 59 kB    00:00
-----
Total                                                                    11 MB/s | 2.4 MB   00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :

```

```

Installing : httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch                8/13
Installing : httpd-core-2.4.68-1.amzn2023.0.1.x86_64                    9/13
Installing : mod_http2-2.0.42-1.amzn2023.0.1.x86_64                    10/13
Installing : mod_lua-2.4.68-1.amzn2023.0.1.x86_64                      11/13
Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch          12/13
Installing : httpd-2.4.68-1.amzn2023.0.1.x86_64                        13/13
Running scriptlet: httpd-2.4.68-1.amzn2023.0.1.x86_64                  13/13
Verifying : apr-1.7.5-1.amzn2023.0.4.x86_64                            1/13
Verifying : apr-util-1.6.5-1.amzn2023.0.1.x86_64                       2/13
Verifying : apr-util-ldb-1.6.5-1.amzn2023.0.1.x86_64                   3/13
Verifying : apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64               4/13
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch          5/13
Verifying : httpd-2.4.68-1.amzn2023.0.1.x86_64                         6/13
Verifying : httpd-core-2.4.68-1.amzn2023.0.1.x86_64                   7/13
Verifying : httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch              8/13
Verifying : httpd-tools-2.4.68-1.amzn2023.0.1.x86_64                   9/13
Verifying : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                     10/13
Verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch                       11/13
Verifying : mod_http2-2.0.42-1.amzn2023.0.1.x86_64                    12/13
Verifying : mod_lua-2.4.68-1.amzn2023.0.1.x86_64                      13/13

Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64          apr-util-1.6.5-1.amzn2023.0.1.x86_64      apr-util-ldb-1.6.5-1.amzn2023.0.1.x86_64      apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch  httpd-2.4.68-1.amzn2023.0.1.x86_64      httpd-core-2.4.68-1.amzn2023.0.1.x86_64      httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch
  httpd-tools-2.4.68-1.amzn2023.0.1.x86_64      libbrotli-1.0.9-4.amzn2023.0.2.x86_64      mailcap-2.1.49-3.amzn2023.0.3.noarch      mod_http2-2.0.42-1.amzn2023.0.1.x86_64
  mod_lua-2.4.68-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-10-0-2-206 ~]$ sudo systemctl start httpd
[ec2-user@ip-10-0-2-206 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-10-0-2-206 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-09-16 10:52:38 UTC; 16s ago
     Docs: man:httpd.service(8)
  Main PID: 21881 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec 0; Bytes served/sec: 0 B/sec"
     Tasks: 177 (limit: 1059)
    Memory: 14.0M
       CPU: 85ms
  CGroup: /system.slice/httpd.service
          └─21881 /usr/sbin/httpd -DFOREGROUND
            └─21962 /usr/sbin/httpd -DFOREGROUND
              └─21968 /usr/sbin/httpd -DFOREGROUND
                └─21969 /usr/sbin/httpd -DFOREGROUND
                  └─21975 /usr/sbin/httpd -DFOREGROUND

Sep 16 10:52:38 ip-10-0-2-206.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Sep 16 10:52:38 ip-10-0-2-206.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Sep 16 10:52:38 ip-10-0-2-206.ec2.internal httpd[21881]: Server configured, listening on: port 80
[ec2-user@ip-10-0-2-206 ~]$ echo "hi$hello from Private Apache Server</h1>" | sudo tee /var/www/html/index.html
hi$hello from Private Apache Server</h1>
[ec2-user@ip-10-0-2-206 ~]$ curl http://localhost
hi$hello from Private Apache Server</h1>
[ec2-user@ip-10-0-2-206 ~]$ sudo systemctl is-active httpd
active
[ec2-user@ip-10-0-2-206 ~]$ |

```

8. Create one application load balancer and attach it to both EC2 instances.

Successfully created the target group: **Autoscaling-TG**. Anomaly detection is automatically applied to all registered targets. Results can be viewed in the Targets tab.

Autoscaling-TG

Details

arn:aws:elasticloadbalancing:us-east-1:956035085469:targetgroup/Autoscaling-TG/ac3e64f1c1548740

Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1	VPC vpc-02bf85d4183f5fd76
IP address type IPv4	Load balancer None associated		

2 Total targets	0 Healthy	0 Unhealthy	2 Unused	0 Initial	0 Draining
--------------------	--------------	----------------	-------------	--------------	---------------

0 Anomalous

Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.

[Targets](#) | [Monitoring](#) | [Health checks](#) | [Attributes](#) | [Tags](#)

Successfully created load balancer: **Autoscaling-ALB**
It might take a few minutes for your load balancer to fully set up and route traffic. Targets will also take a few minutes to complete the registration process and pass initial health checks.

Introducing new Capacity Utilization metric for ALB
Monitor how much of your load balancer's available capacity was used to meet traffic demand. Visit the [Capacity](#) or [Monitoring](#) tabs to view utilization data.

Autoscaling-ALB

Details

Load balancer type Application	Status Provisioning	VPC vpc-02bf85d4183f5fd76	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z355XDOTRQ7X7K	Availability Zones subnet-05a0b31a7b215c8af us-east-1b (use1-az2) subnet-09f27cca8d5b6feda us-east-1a (use1-az1)	Date created September 16, 2026, 16:31 (UTC+05:30)

Load balancer ARN
arn:aws:elasticloadbalancing:us-east-1:956035085469:loadbalancer/app/Autoscaling-ALB/117d94f8dcf27494

DNS name
[Autoscaling-ALB-176424503.us-east-1.elb.amazonaws.com \(A Record\)](#)

9. Store application load balancer logs in S3.

Successfully created bucket "autoscaling-alb-logs-poojith-2026"

autoscaling-alb-logs-poojith-2026

[Objects](#) | [Metadata](#) | [Properties](#) | [Permissions](#) | [Metrics](#) | [Management](#) | [File systems](#) | [Access Points](#)

Objects (0)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name	Type	Last modified	Size	Storage class
No objects You don't have any objects in this bucket.				

[Upload](#)

aws [Search] [Alt+S] United States (N. Virginia) Poojith R (956035085469) Poojith R

EC2 > Load balancers > Autoscaling-ALB

Capacity reservations Capacity Manager New

▼ Images
AMIs
AMI Catalog

▼ Elastic Block Store
Volumes
Snapshots
Lifecycle Manager

▼ Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

▼ Load Balancing
Load Balancers
Target Groups
Trust Stores

▼ Auto Scaling
Auto Scaling Groups

Settings

Successfully modified load balancer attributes.

Introducing new Capacity Utilization metric for ALB
Monitor how much of your load balancer's available capacity was used to meet traffic demand. Visit the [Capacity](#) or [Monitoring](#) tabs to view utilization data.

Autoscaling-ALB

Details

Load balancer type Application	Status Active	VPC vpc-02bf85d4183f5fd76	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z35SXDOTRQ7X7K	Availability Zones subnet-05a0b31a7b215c8af us-east-1b (use1-az2) subnet-09f27cca8d5b6feda us-east-1a (use1-az1)	Date created September 16, 2026, 16:31 (UTC+05:30)
Load balancer ARN arn:aws:elasticloadbalancing:us-east-1:956035085469:loadbalancer/app/Autoscaling-ALB/117d94f8dcf27494		DNS name info Autoscaling-ALB-176424503.us-east-1.elb.amazonaws.com (A Record)	

aws [Search] [Alt+S] United States (N. Virginia) Poojith R (956035085469) Poojith R

EC2 > Load balancers > Autoscaling-ALB

Capacity reservations Capacity Manager New

▼ Images
AMIs
AMI Catalog

▼ Elastic Block Store
Volumes
Snapshots
Lifecycle Manager

▼ Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

▼ Load Balancing
Load Balancers
Target Groups
Trust Stores

▼ Auto Scaling
Auto Scaling Groups

Settings

Off

Availability Zone routing configuration

Cross-zone load balancing On	ARC zonal shift integration Disabled
--	--

Protection

Deletion protection Off

Logging

Access logs - legacy S3 location: autoscaling-alb-logs-poojith-2026	Connection logs - legacy Off	Health check logs - legacy Off
---	--	--

Logging options
Log delivery to CloudWatch, Kinesis Firehose, and S3 is configurable through an integration with Logging - by Amazon CloudWatch. Visit [Logging](#) in the [Integrations](#) tab.

autoscaling-alb-176424503.us-east-1.elb.amazonaws.com

Not secure autoscaling-alb-176424503.us-east-1.elb.amazonaws.com

Hello from Private Apache Server

16/

Amazon S3 Buckets autoscaling-alb-logs-poojith-2026 AWSLogs/ 956035085469/ elasticloadbalancing/ us-east-1/ 2026/ 09/ 16/

Objects (1) Copy S3 URI Copy URL Download Open L^A Delete Actions Create folder Upload

Find objects by prefix

Name	Type	Last modified	Size	Storage class
956035085469_elasticloadbalancing_us-east-1_app.Autoscaling-ALB.117d94f8dcf27494_20260916T1210Z_54.156.23.74_67jn52hx.log.gz	gz	September 16, 2026, 17:40:03 (UTC+05:30)	329.0 B	Standard

10. Store the VPC flow logs in a CloudWatch log group.

CloudWatch Log management

Log group "VPC/Autoscaling-VPC-FlowLogs" has been created.

Log groups Data sources - new Summary

Get-set-go with unified data store

Log groups (3) Filters Actions View in Logs Insights Start tailing Create log group

By default, we only load up to 10,000 log groups.

Filter log groups or try pattern search Exact match

Log group	Log class	Anomaly d...	Deleti...	Bearer...	Dat...	Sen...	Retenti...
/VPC/Autoscaling-VPC-FlowLogs	Standard	Configure	Off	Off	-	-	Never exp...
aws-cloudtrail-logs-956035085469-440e7900	Standard	Configure	Off	Off	-	-	Never exp...
my-cloudwatch-log-group1	Standard	Configure	Off	Off	-	-	Never exp...

VPC dashboard VPC Your VPCs fl-01a75e741d3f65c74

Filter by VPC:

AWS Global View

Virtual private cloud Your VPCs Subnets Route tables Internet gateways Egress-only Internet gateways Carrier gateways DHCP option sets Elastic IPs Managed prefix lists NAT gateways Peering connections Route servers

Security Network ACLs

Successfully created flow log for the following resource: vpc-02bf85d4183f5fd76

fl-01a75e741d3f65c74 / Autoscaling-VPC-FlowLog Actions

Details

Flow log ID fl-01a75e741d3f65c74	Destination Type cloud-watch-logs	Traffic Type All	File Format -
Name Autoscaling-VPC-FlowLog	Destination Name /VPC/Autoscaling-VPC-FlowLogs	Max Aggregation Interval 1 minute	Hive Compatible Partitions -
State Active	IAM Role arn:aws:iam::956035085469:role/poojit	Log Format Default	Partition Logs -
Creation Time Wednesday, 16 September 2026 at 17:52:02 GMT+5:30	Cross Account IAM Role -	Tag field specifications -	

Tags Integrations

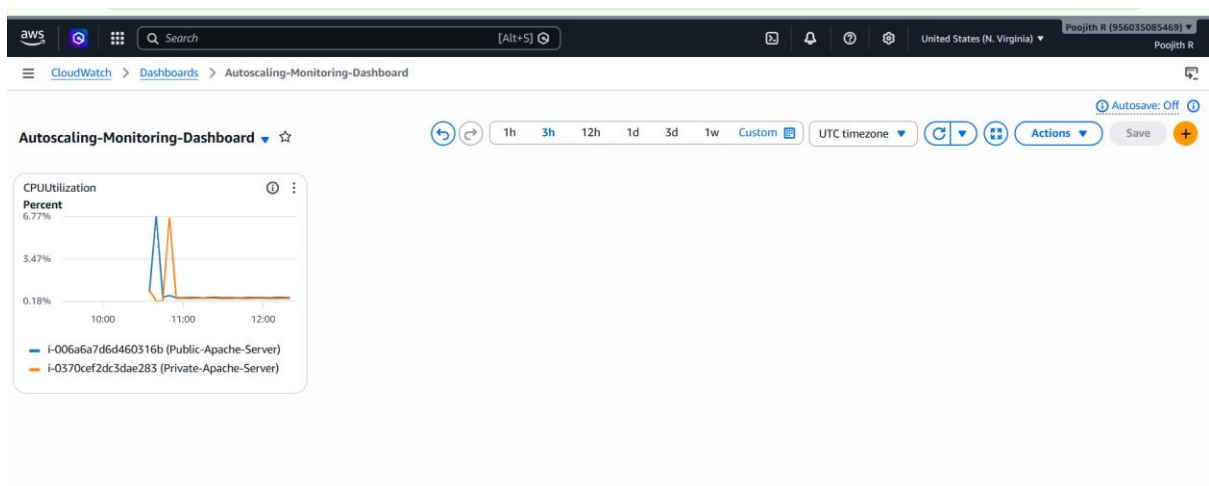
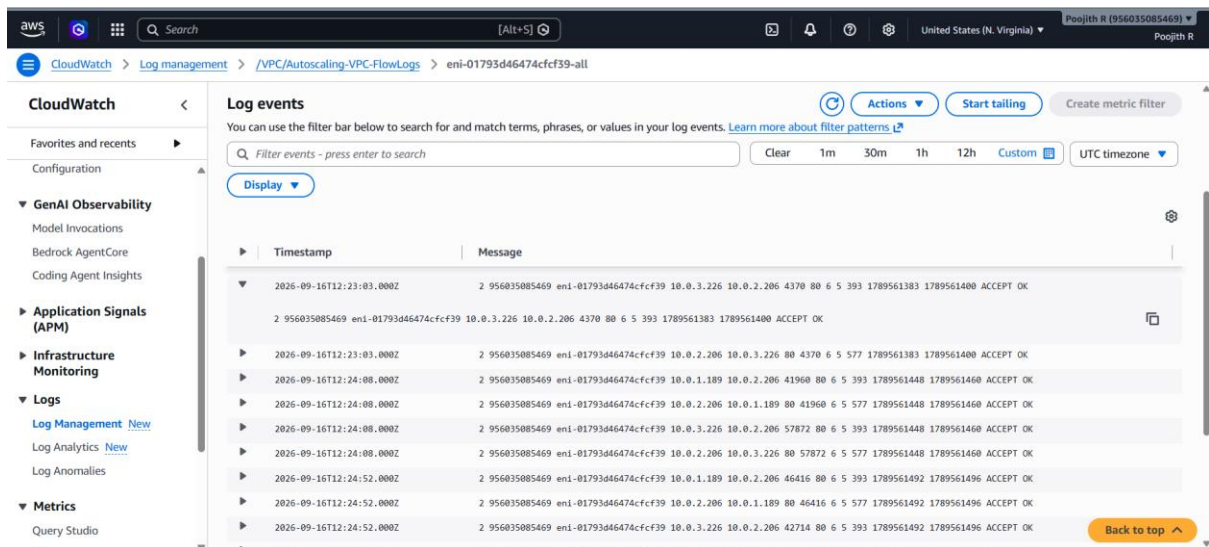
Tags

Search tags

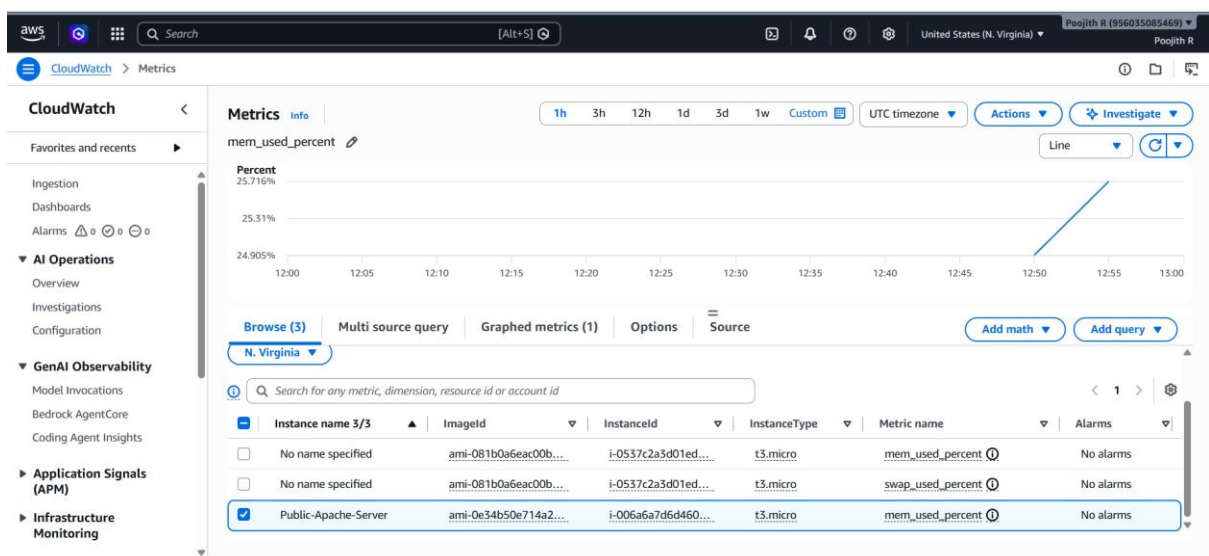
Key Value

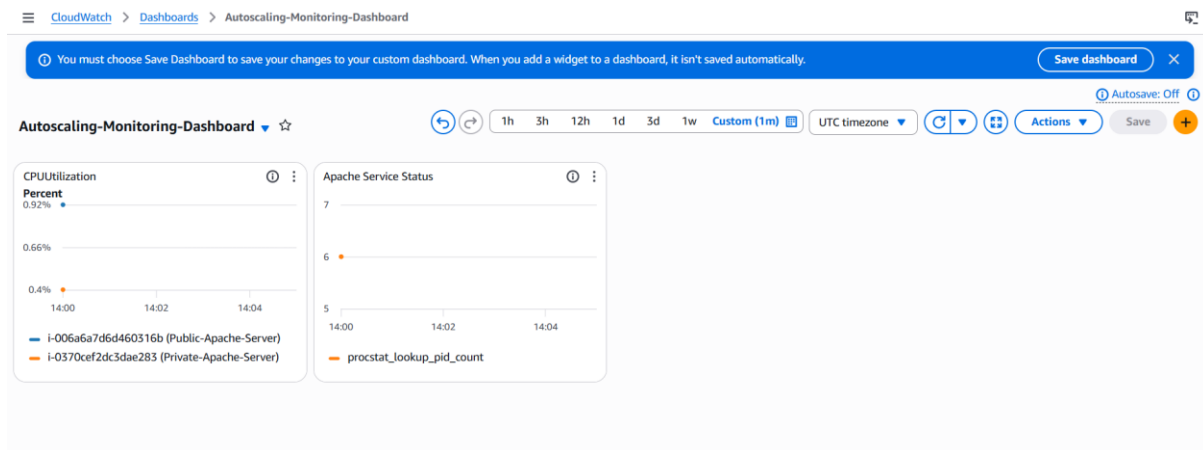
Manage tags

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11. Create monitoring dashboards to monitor CPU utilization and to monitor the Apache service.





12. If CPU utilization is more than 70%, then it should trigger auto scaling and launch new instance.

Implementation

- Create an **EC2 Launch Template**.
- Select Amazon Linux 2023.
- Select the required instance type.
- Configure the Apache security group.
- Add User Data to automatically install and start Apache.
- Create an **Auto Scaling Group**.
- Select the Launch Template.
- Select the VPC.
- Select **private subnets in multiple AZs**.
- Set:
 - Minimum capacity: 1
 - Desired capacity: 2
 - Maximum capacity: 5
- Attach the Apache target group to the ASG.
- Enable ELB health checks.
- Create a **dynamic scaling policy**.
- Select CPU utilization as the scaling metric.

- Set the target/threshold to approximately **70%**, according to the assignment.
- Configure the scaling policy to increase desired capacity by 2 when the CPU condition is met.
- CloudWatch monitors CPU utilization.
- When CPU exceeds the configured threshold, the alarm triggers the scaling policy.
- Auto Scaling launches a new EC2 instance.
- The new instance automatically installs Apache through User Data.
- The new instance registers with the ALB target group.

The screenshot shows the AWS Management Console interface for the 'web-server-template' Launch Template. The left sidebar contains navigation links for EC2, Launch templates, and various EC2 services. The main content area displays the 'Launch template details' for the template with ID 'lt-08be0c285deed5688'. Below this, the 'Launch template version details' are shown for version 1 (Default), including the AMI ID 'ami-0354c98ae10b02961' and instance type 't3.micro'. The 'Date created' is listed as '2026-09-11T11:55:45.000Z'. The 'Actions' button is visible in the top right corner of the version details section.

The screenshot shows the 'Create Auto Scaling group' wizard in the AWS Management Console. The 'VPC' section is selected, showing the chosen VPC 'vpc-06c94a95d4274b209 (my-vpc-virginia)'. The 'Availability Zones and subnets' section is also visible, showing the selected subnets 'use1-az1 (us-east-1a) | subnet-09403bfff77bd60066 (private-subnet)'. The 'Availability Zone distribution' section is partially visible at the bottom, indicating that Auto Scaling will automatically balance instances across Availability Zones.

EC2

>

Auto Scaling groups

>

Create Auto Scaling group

Step 4 - optional

Configure group size and scaling

Step 5 - optional

Add notifications

Step 6 - optional

Add tags

Step 7

Review

Scaling group name

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity

Specify your group size.

2

Scaling

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

1

Equal or less than desired capacity

Max desired capacity

5

Equal or greater than desired capacity

EC2

>

Auto Scaling groups

>

Create Auto Scaling group

Scaling policy name

Target Tracking Policy

Metric type

Info

Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

Average CPU utilization

Target value

70

Instance warmup

Info

300

seconds

☐

Disable scale in to create only a scale-out policy

Hello from Auto Scaling EC2

Apache HTTP Server is running successfully.

This instance was launched by the Auto Scaling Group.

